Vector Calculus and Complex Analysis Classwork (2025/2/24)

Name:

Student ID:

Consider the vector field $\boldsymbol{F}(x,y)=\left(\frac{x^2-y^2}{(x^2+y^2)^2},\frac{2xy}{(x^2+y^2)^2},0\right)$ on $\mathbb{R}^2\setminus\{(0,0)\}$. Find curl($\boldsymbol{F}$). Is it possible to find a real-value function $f(x, y)$ such that $\boldsymbol{F}(x,y)=\nabla f$? If yes, find $f(x, y)$. If no, explain why not.

Solution

$$\boldsymbol{F}(x,y)=\left(\frac{x^2-y^2}{(x^2+y^2)^2},\frac{2xy}{(x^2+y^2)^2}\right)$$

$$\frac{\partial}{\partial y}\frac{x^2-y^2}{(x^2+y^2)^2}=\frac{-(x^2+y^2)^2 2y-(x^2-y^2)2(x^2+y^2)2y}{(x^2+y^2)^4}=-2y\frac{(x^2+y^2)+2(x^2-y^2)}{(x^2+y^2)^3}=-2y\frac{3x^2-y^2}{(x^2+y^2)^3}$$

$$\frac{\partial}{\partial x}\frac{2xy}{(x^2+y^2)^2}=\frac{(x^2+y^2)^2 2y-2xy\cdot 2(x^2+y^2)2x}{(x^2+y^2)^4}=2y\frac{(x^2+y^2)-4x^2}{(x^2+y^2)^3}=2y\frac{-3x^2+y^2}{(x^2+y^2)^4}=\frac{\partial}{\partial y}\frac{x^2-y^2}{(x^2+y^2)^2}$$

$\text{curl}(\boldsymbol{F})=(0,0,0)$.

Yes. We can find a real-value function $f(x,y)$ such that $\boldsymbol{F}(x,y)=\nabla f$.

$$\frac{\partial f}{\partial x}=\frac{x^2-y^2}{(x^2+y^2)^2},\quad \frac{\partial f}{\partial y}=\frac{2xy}{(x^2+y^2)^2}$$

$$f(x,y)=\int\frac{2xy}{(x^2+y^2)^2}dy=\int\frac{x}{(x^2+y^2)^2}d(x^2+y^2)=\frac{-x}{x^2+y^2}+g(x)$$

$$\frac{\partial f}{\partial x}=\frac{-(x^2+y^2)+x(2x)}{(x^2+y^2)^2}+g'(x)=\frac{x^2-y^2}{(x^2+y^2)^2}+g'(x)$$

$g'(x)=0$

Choose $g(x)=0$ and $f(x,y)=\dfrac{-x}{x^2+y^2}$ on $\mathbb{R}^2\setminus\{(0,0)\}$.

"No. It is not possible because $\mathbb{R}^2\setminus\{(0,0)\}$ is not simply connected."

Such argument is incorrect.

"curl($\boldsymbol{F}$) = **0** and D is simply connected" $\Rightarrow$ there must be a real-value function $f(x,y)$ such that $\boldsymbol{F}(x,y)=\nabla f$.

If "curl($\boldsymbol{F}$) = **0** and D is not simply connected", then it might be possible/might not possible to find a real-value function $f(x,y)$ such that $\boldsymbol{F}(x,y)=\nabla f$.

It does not mean that it must not possible find a real-value function $f(x,y)$ such that $\boldsymbol{F}(x,y)=\nabla f$.